ANALYTICAL CUMULATIVE EXAM
April 5, 1996
Theme: Dissolution of CO_2 in water

(a) (20 pts) Dissolved carbon dioxide forms a diprotic acid with water. Write down the two chemical equations for the acid dissociations of that diprotic acid with water. Also, write the mathematical expressions relating the acid dissociation constants to concentrations. Let the first acid dissociation constant be $\mathbf{Ka_1 = 4.45 \times 10^{-7}}$, and the second be $\mathbf{Ka_2 = 4.69 \times 10^{-11}}$.

(b) (10 pts) The diprotic acid's behavior appears anomalous at first, because the **value of Ka_1 is about 10^2 to 10^4 times smaller** than the acid dissociation constants for other carboxylic acids. The reason for this seeming anomaly is not that H_2CO_3 is unusual, but rather, that the value commonly given for Ka_1 applies to the chemical equation:

$$\text{all dissolved } CO_2 \rightleftharpoons HCO_3^- + H^+$$

where the "all dissolved CO_2" consists of both $CO_2(aq)$ and the diprotic acid.

Derive a new mathematical expression for Ka_1, taking into consideration the above equation, so that it will contain both $CO_2(aq)$ and the diprotic acid species.

Now that you have this expression, consider the fact that **only 0.2% of "all dissolved CO_2" is in the diprotic acid form**. Calculate the value of the equilibrium constant for the *true equilibrium* of the diprotic acid in water (when the true value of the diprotic acid is used instead of the value for "all dissolved CO_2"). How does this new acid dissociation constant now compare with those for other carboxylic acids?

(c) (20 pts) Derive the complete expression (below) for $[H_3O^+]$ as a function of the pressure of CO_2 (P_{CO2}) in the atmosphere above water, the acid dissociation constants, Ka_1 and Ka_2, the equilibrium constant for the solubility of CO_2 in water, K, and the water dissociation constant, K_w.

$$[H_3O^+]^3 - [H_3O^+](K\,Ka_1\,P_{CO_2} + K_w) - 2\,K\,Ka_1\,Ka_2\,P_{CO_2} = 0$$

Take into account the answers from (a) and (b) above. Include the dissociation of water (equilibrium constant $\mathbf{K_w = 10^{-14}}$). Don't forget to consider charge and/or mass balance equations. Also, note that $CO_2(g)$ dissolves in water with the following chemical equilibrium, and with the equilibrium constant of $\mathbf{K = 10^{-1.5}}$.

$$CO_{2\,(g)} + H_2O_{(l)} \rightleftharpoons \text{all dissolved } CO_2$$

(d) (20 pts) What is the pH of water in equilibrium with an atmosphere having a pressure of $\mathbf{P_{CO2} = 10^{-3.5}}$ **atm**? State any approximations that you make. It is possible to make two assumptions to simplify the mathematics.

(e) (10 pts) Carbonated beverages are created by equilibrating a drink with a high pressure of CO_2. What pressure must be above the drink to attain a **pH of 4.9**?

(f) (20 pts) When salt (NaCl) is added to a freshly opened container of a softdrink, the drink fizzes and foams. When salt is added to a softdrink which has been exposed to air for a long time, there is no signficant fizzing or foaming. Explain these phenomena.